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Course, Year & Section: BSIT 4B

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Week 6 Laboratory Guide

Network Configuration and Management

Part 1 – Current Network Configuration (Baseline)

Active Interfaces (ip a):

```
root@ubuntu-server:/var/snap/subiquity/7225# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:87:f3:e0 brd ff:ff:ff:ff:ff:ff
    altname enx08002787f3e0
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86252sec preferred_lft 86252sec
    inet6 fd00::a00:27ff:fe87:f3e0/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86253sec preferred_lft 14253sec
    inet6 fe80::a00:27ff:fe87:f3e0/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
root@ubuntu-server:/var/snap/subiquity/7225#
```

Default Route & DNS (ip route / cat /etc/resolv.conf):

```
root@ubuntu-server:/var/snap/subiquity/7225# df
Filesystem      1K-blocks    Used Available Use% Mounted on
tmpfs            812456      1152     811304   1% /run
/dev/sr0         2844572 2844572        0 100% /cdrom
/cow             4062264    76544   3985720   2% /
tmpfs            4062264        0   4062264   0% /dev/shm
none             1024        0       1024   0% /run/credentials/systemd-journald.service
none             1024        0       1024   0% /run/credentials/systemd-resolved.service
tmpfs            4062264        0   4062264   0% /tmp
none             1024        0       1024   0% /run/credentials/systemd-networkd.service
/dev/sdb1       10217748    2072   9675056   1% /projectdata
root@ubuntu-server:/var/snap/subiquity/7225#
```

Part 2 – Server A Static IP Configuration (192.168.1.150)

Netplan Configuration File (/etc/netplan/00-installer-config.yaml):

```
File Machine View Input Devices Help
dhcp4: true
dhcp6: true
accept-ra: true
zz-all-eth:
  match:
    name: "eth*"
  dhcp4: true
  dhcp6: true
  accept-ra: true
```

Applied IP Verification (ip a):

```

** (process:2784): WARNING **: 11:09:27.609: `gateway4` has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.

** (process:2784): WARNING **: 11:09:27.739: `gateway4` has been deprecated, use default routes instead.
See the 'Default routes' section of the documentation for more details.
root@ubuntu-server:/var/snap/subiquity/7225# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:87:f3:e0 brd ff:ff:ff:ff:ff:ff
    altname enx08002787f3e0
    inet 192.168.1.150/24 brd 192.168.1.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86374sec preferred_lft 86374sec
    inet6 fd00::a00:27ff:fe87:f3e0/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86375sec preferred_lft 14375sec
    inet6 fe80::a00:27ff:fe87:f3e0/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
root@ubuntu-server:/var/snap/subiquity/7225# _

```

Part 3 – Server B DHCP Configuration
Netplan Configuration File (/etc/netplan/00-installer-config.yaml):

```

network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: yes

```

Dynamic IP Verification (ip

a):

```

root@ubuntu-server:/var/snap/subiquity/7225# sudo netplan apply
root@ubuntu-server:/var/snap/subiquity/7225# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:87:f3:e0 brd ff:ff:ff:ff:ff:ff
    altname enx08002787f3e0
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86397sec preferred_lft 86397sec
    inet6 fd00::a00:27ff:fe87:f3e0/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86398sec preferred_lft 14398sec
    inet6 fe80::a00:27ff:fe87:f3e0/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
root@ubuntu-server:/var/snap/subiquity/7225# _

```

Part 4 – Troubleshooting & Connectivity Verification
Ping Results (ping google.com / Server-to-Server):

```

    valid_lft forever preferred_lft forever
root@ubuntu-server:/var/snap/subiquity/7225# ping -c 4 google.com
PING google.com (172.217.27.78) 56(84) bytes of data:
64 bytes from nrt12s15-in-f14.1e100.net (172.217.27.78): icmp_seq=1 ttl=255 time=16.8 ms
64 bytes from nrt12s15-in-f14.1e100.net (172.217.27.78): icmp_seq=2 ttl=255 time=17.3 ms
64 bytes from nrt12s15-in-f14.1e100.net (172.217.27.78): icmp_seq=3 ttl=255 time=17.3 ms
64 bytes from nrt12s15-in-f14.1e100.net (172.217.27.78): icmp_seq=4 ttl=255 time=17.3 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3028ms
rtt min/avg/max/mdev = 16.806/17.179/17.327/0.216 ms
root@ubuntu-server:/var/snap/subiquity/7225# _

```

Traceroute Results (traceroute google.com):

```
o VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ubuntu-server:/var/snap/subiquity/7225# traceroute google.com
raceroute to google.com (172.217.27.78), 30 hops max, 60 byte packets
1  10.0.2.2 (10.0.2.2)  0.404 ms  0.311 ms  0.276 ms
2  * * *
3  * * *
4  * * *
5  * * *
6  * * *
7  * * *
8  * * *
9  * * *
0  * * *
1  * * *
2  * * *
3  * * *
4  * * *
5  * * *
6  * * *
7  * * *
8  * * *
9  * * *
0  * * *
1  * * *
2  * * *
3  * * *
4  * * *
5  * * *
6  * * *
7  * * *
8  * * *
9  * * *
0  * * *
root@ubuntu-server:/var/snap/subiquity/7225#
```

DNS Resolution Test (nslookup google.com):

```
/etc/netplan/00-installer-config.yaml      mkdir
root@ubuntu-server:/var/snap/subiquity/7225#
root@ubuntu-server:/var/snap/subiquity/7225# nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 172.217.27.78
Name:   google.com
Address: 2404:6800:4017:807::200e

root@ubuntu-server:/var/snap/subiquity/7225#
```

Active Connections (ss -tuln):

```
root@ubuntu-server:/var/snap/subiquity/7225# ss -tuln
Netid      State      Recv-Q     Send-Q     Local Address:Port
udp        UNCONN     0           0           127.0.0.54:53
udp        UNCONN     0           0           127.0.0.53%lo:53
udp        UNCONN     0           0           10.0.2.15%enp0s3:68
udp        UNCONN     0           0           127.0.0.1:323
udp        UNCONN     0           0           [::1]:323
tcp        LISTEN     0           4096        127.0.0.54:53
tcp        LISTEN     0           4096        127.0.0.53%lo:53
tcp        LISTEN     0           4096        0.0.0.0:22
tcp        LISTEN     0           4096        [::]:22
root@ubuntu-server:/var/snap/subiquity/7225#
```

Part 5 – Reflection

1. What challenges did you encounter when modifying network configurations via Netplan?

Configuring Netplan requires strict adherence to YAML syntax, where proper indentation and spacing are critical. A single misplaced space can cause netplan apply to fail, locking you out or disrupting network connectivity until corrected.

2. How do static IPs and DHCP differ in terms of network administration use cases?

Static IPs are essential for servers hosting internal services (like databases, web servers, or DNS) because their addresses must remain constant for reliable client access. DHCP, on the other hand, is ideal for client devices, workstations, and flexible environments where dynamic address allocation prevents IP conflicts and reduces administrative overhead.